

Force joins the token zoo late.

π_0 parent $\rightarrow$ late 6D force token $\rightarrow$ FVLMoE (4 experts, top-1) $\rightarrow$ additive FM guidance. Association cortex for force; π_0 's FM head stays motor cortex.

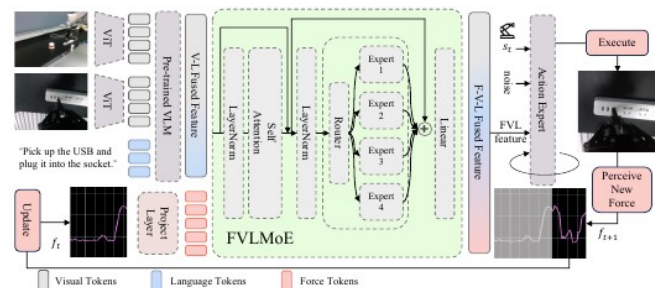


Figure 3: Overview of our ForceVLA model. Visual and language inputs are processed by a pre-trained VLM to form contextual embeddings. External force signals are projected and fused with VLM outputs via the FVLMoE module. The resulting multimodal features guide a flow-based action head to generate contact-aware robot actions.

that are used to select a small subset of k experts (typically $k = 1$ or $k = 2$, where $k \ll N$) from the total pool of N experts. The input token x is then routed only to these k active experts. The outputs of these active experts, $E_i(x)$, are subsequently aggregated, commonly through a weighted sum where the weights $g_i(x)$ are also derived from the gating network. The final output $y(x)$ of the MoE layer can be expressed as: $y(x) = \sum_{i \in \text{TopK}(G(x))} g_i(x) E_i(x)$, where $\text{TopK}(G(x))$ denotes the set of indices of the top- k experts selected by the gating network for input x .

4 ForceVLA

Five facts that do not change with the seat

Late 6-axis F/T + FVLMoE

01

After VLM (PaliGemma/SigLIP): project wrench → force token → FVLMoE (E=4, top-1) → additive guide into π_0 flow-matching action head. Force stays late.

Contact-rich on Flexiv

02

Insert USB / plug, pump bottle, wipe board, peel cucumber. ForceVLA-Data: 244 traj / ~140k steps / ~50 demos per task. Quest3 teleop.

Must-cites

03

Avg 60.5 vs π_0 -base w/o F 37.3 (+23.2). Naive w/ F only 40.2 (+2.9). Early MoE-before-VLM 0% → late FVLMoE 80% (Tab.3). Occlusion 90%. FAST+naive force 31.0→14.2.

CLEAR contrasts

04

NOT MoPA-PD ancestor (MoPA = MP privilege→vision student; ForceVLA = deploy force into VLA). NOT JEPA/WM child. No $\pi_{0.5}$ bake-off — do not invent.

Number hygiene

05

Typo fix: 37.3 = w/o F (not w/ F). Preserve paper numbers. No invented BEHAVIOR SR. Pump caveat: naive w/F 83 > ForceVLA 67 — do not overclaim.

π_0 parent $\rightarrow$ late force $\rightarrow$ FVLMoE $\rightarrow$ FM

1

PaliGemma / SigLIP VLM

Vision + language contextual embeddings.
Pretrained geometry to protect.

2

π_0 FM action expert

Conditional flow matching + action chunks.
Vision+language+proprio parent.

3

Late 6-axis force token

$\varphi_F: \mathbb{R}^6 \rightarrow 2048$ AFTER VLM. Not early into
VL prefix.

4

FVLMoE E=4 top-1

Attn refine $\rightarrow$ sparse MoE $\rightarrow$ residual $\rightarrow$ project
2048 $\rightarrow$ 1024.

5

Additive FM suffix

G_{FVLMoE} added to proprio + noisy-action
features each denoising step.

6

ForceVLA-Data + Flexiv

244 traj / 140k / ~50/task. Contact-rich real
policy.

What ForceVLA cites as parents and neighbors

π_0 Black et al.

Direct architectural parent. PaliGemma + FM action expert. Baselines π_0 -base / π_0 -fast $\pm$ force.

PaliGemma / SigLIP

Beyer et al.; Zhai et al. VLM backbone for VL contextual embeddings.

Flow matching

Lipman et al.; Liu rectified flow. Action expert family retained.

FAST Pertsch et al.

Baseline action-tokenization foil (π_0 -fast). Not ForceVLA's head.

VLA lineage

OpenVLA / RT-1 / RT-2 / Octo / OXE / DROID / Bridge — related work. Numbers foil is π_0 .

MoE lineage

Shazeer MoE, Switch, GShard, V-MoE, ST-MoE, LIMOE; robotics MoE (MORE, ChatVLA).

Same Alt row. Opposite arrow. Not an ancestor.

MoPA-PD (2111.06383)

Train-time privilege

Classical MP (RRT-Connect) inside MoPA-RL → visual BC → asymmetric SAC.

Deploy

Planner-free pixels. Geometry privilege removed at ship time.

Syllabus

Prior Alt approaches row. Privileged teacher → vision student.

ForceVLA (2505.22159)

Deploy-time modality

Add 6-axis F/T into a π_0 -class VLA via late MoE fusion.

Deploy

Force stream stays at inference. FM action head retained.

Syllabus

LAST Alt row. Complementary question: missing modality for contact-rich VLA.

Poles and absences — do not invent links

JEPA / LeWM / Cosmos

CLEAR contrast poles only. ForceVLA is not a world-model paper. No latent prediction claim.

$\Pi_{0.5}$ (2504.16054)

Cited in related work [21]. ABSENT from experiments. Sibling literature — not measured. Do not invent bake-off.

Π_0 -FAST

Baseline family with/without force. ForceVLA keeps FM. Mythbust: adding force $\neq$ switching FM $\leftrightarrow$ FAST.

Classical impedance

Background of contact-rich control. ForceVLA learns force-conditioned chunks end-to-end rather than hand-tuned compliance loops.

Vision-only VLA meets contact; force corrects

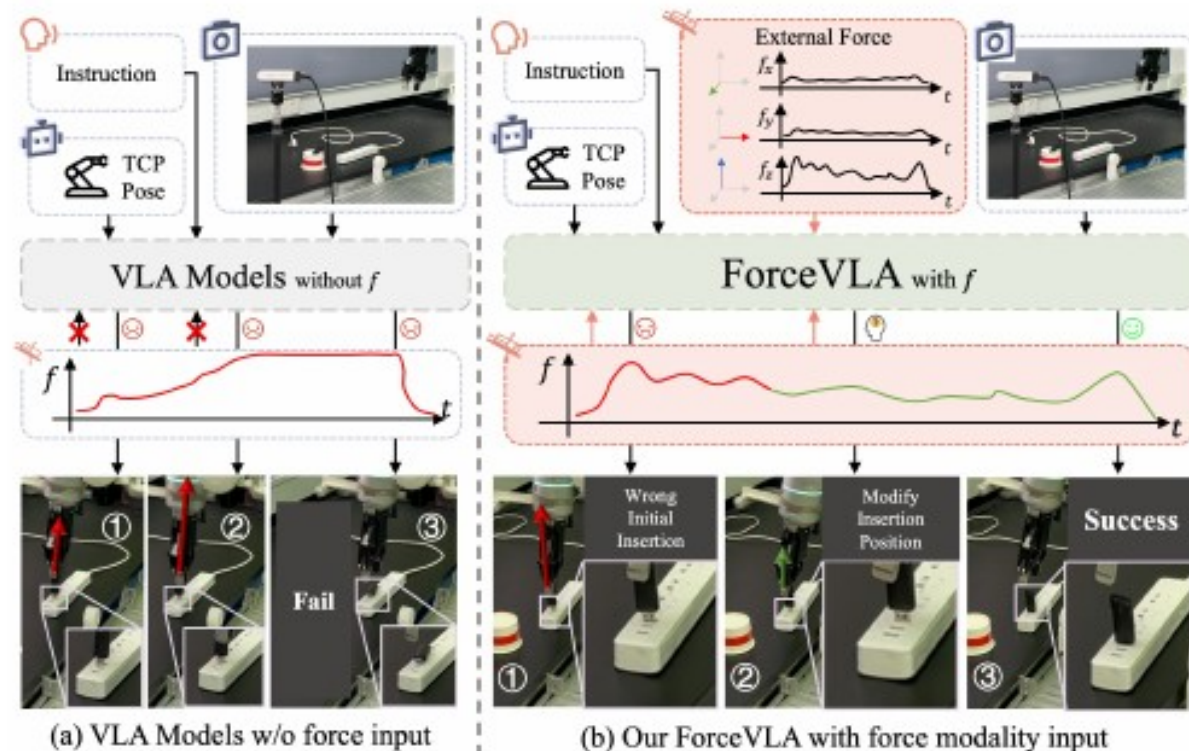


Figure 1: Comparison between ForceVLA and baselines without force input. Without force feedback, the policy fails to correct pose errors and completes insertion incorrectly. In contrast, ForceVLA leverages external force signals to adjust insertion strategies dynamically, leading to successful

What got bolted onto π_0 — exact specs

Force projection

Linear 6 $\rightarrow$ D_VLM = 2048

Pre-MoE encoder

D=2048, N_H=8, D_h=256; MLP expansion 1

MoE

E=4 MLP experts, top-k=1; router 2048 $\rightarrow$ 4

MoE out / action

2048 $\rightarrow$ 1024; action out 1024 $\rightarrow$ 32

State / action proj

32 $\rightarrow$ D_act_e = 1024; τ -MLP 2-layer Swish

Inject

Element-wise add G_FVLMoE into FM suffix

Phase specialization — TENTATIVE labels

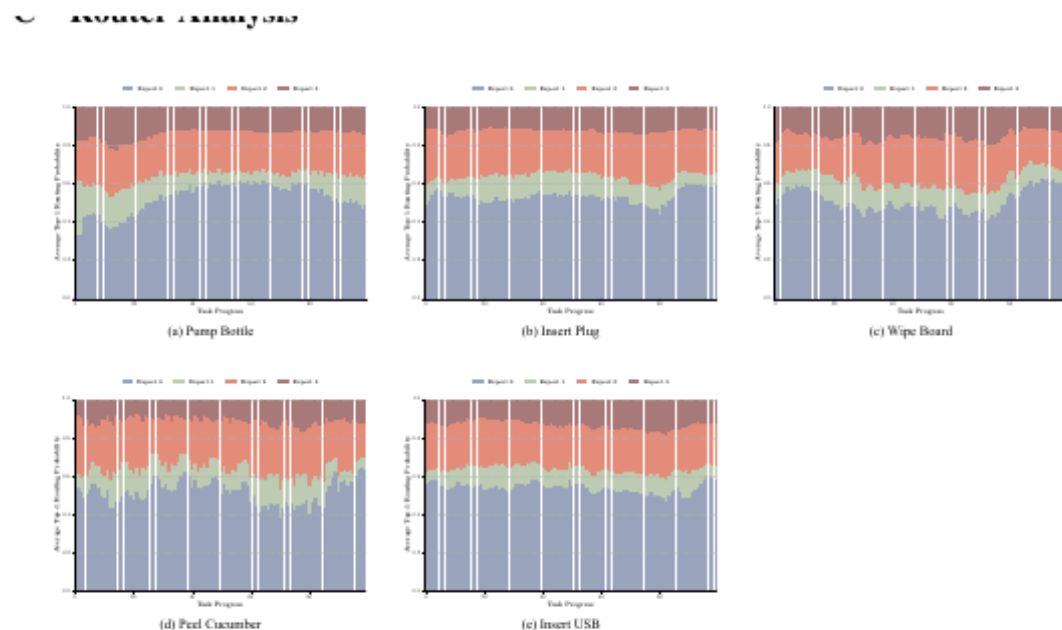


Figure 9: Open-loop evaluation of expert load across different task completion percentages for various tasks: (a) Pump Bottle, (b) Insert Plug, (c) Wipe Board, (d) Peel Cucumber, and (e) Insert USB. Each subplot represents the average expert load (vertical axis) as a function of the task completion percentage (horizontal axis) over the episodes in the test dataset.

To analyze routing dynamics, we first measured the probability distribution over expert selections for each token as it was processed by the router in the MoE module. For variable-length episodes, we applied percentile-based normalization: each task (~ 10 episodes) was processed by segmenting every episode's token sequence into 100 temporally equidistant intervals, computing the mean top-1 probability per segment, and then averaging these means across episodes. This ensured cross-episode comparability while preserving temporal routing dynamics.

Expert 0

~half of tokens. Hypothesized generalist.

Selective

Experts 1/3 more phase-specific on insert/peel.

Wipe

Steadier single-expert preference.

Status

TENTATIVE functional labels. Not proven causal roles. No expert-dropout ablation.

Memorize this sentence

**ForceVLA = π_0 + late 6D force token + FVLMoE
(4 experts, top-1) additively guiding FM chunks;
+23.2 pts avg vs vision-only π_0 -base;
early MoE-into-VLM $\rightarrow$ 0%.**

CLEAR: not MoPA-PD ancestor; not JEPA child; no $\pi_{0.5}$ bake-off.

CLEAR: keeps FM (not FAST); foil is π_0 -base/fast $\pm$ force.

Refuse fabrications

Builds on π_0 FM + PaliGemma	CLEAR
Foil = π_0 -base / π_0 -fast $\pm$ force	CLEAR
Empirical $\pi_{0.5}$ comparison	ABSENT — do not invent
Action head = flow matching	CLEAR
Late $\gg$ early MoE-into-VLM	CLEAR (Tab.3)
MoPA-PD / JEPA ancestry	FALSE — contrast only
ForceVLA-Data 244 / 140k	CLEAR
Expert-0 = generalist role	TENTATIVE
BEHAVIOR official baseline	TENTATIVE / unverified

Late force. Parent is π_0 . Contrasts are labeled.

Association cortex for force. Motor cortex stays FM.

Say the exam bite. Refuse MoPA/JEPA ancestry and invented $\pi_{0.5}$ numbers.

Next: Visionary — BEHAVIOR contact-rich; F/T + late MoE on modern VLA; NO Discord/Drive.